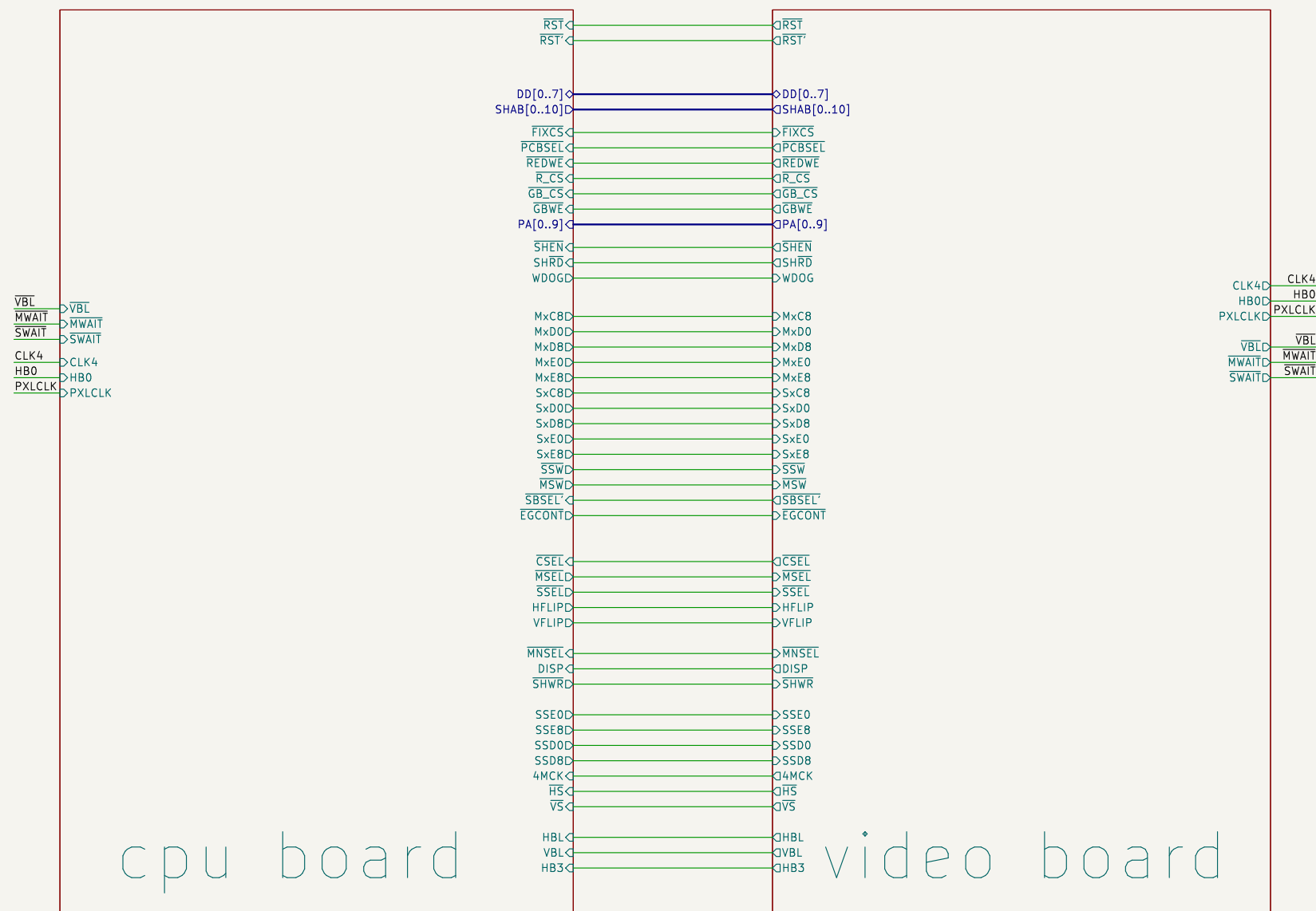




cpu board

video board

```
tehkanwch
bootleg board
```

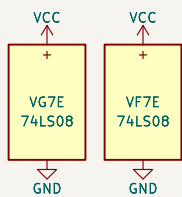
- original PCB has through-hole vias whereas the bootleg has filled vias
 - original uses three custom packages from Mitsubishi with logic inside.
- These were replaced by a small daughter board on the bootleg, serving the same functionality
- different edge connectors
 - debug edge connector on video board not present on bootleg
 - cabinet input logic partly missing and rewired on the bootleg to replace the trackball interface for a joystick one
 - different power amplifier for sound without heat sink on the bootleg
 - slightly different frequencies for crystal oscillators (cost reduction?)
 - device reference markers (silkscreen) on bootleg have reversed order on bootleg

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 Sheet: /
 File: tehkanwch.kicad_sch

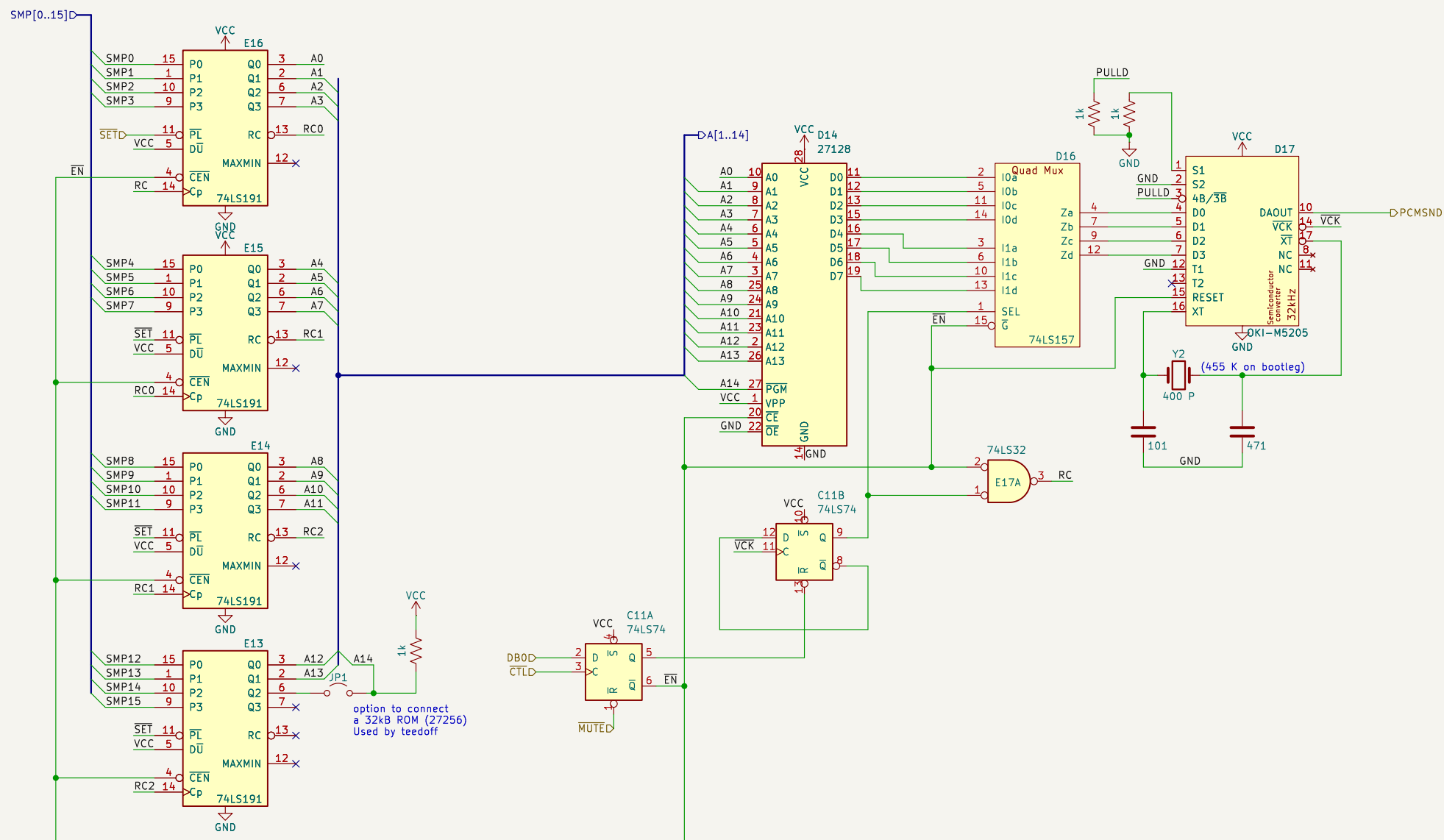


Title: TEHKAN WORLD CUP (tehkanwch bootleg)

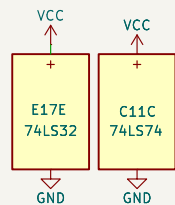
Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 1/26



Rev:
Id: 2/26



option to connect
a 32kB ROM (27256)
Used by teedoff

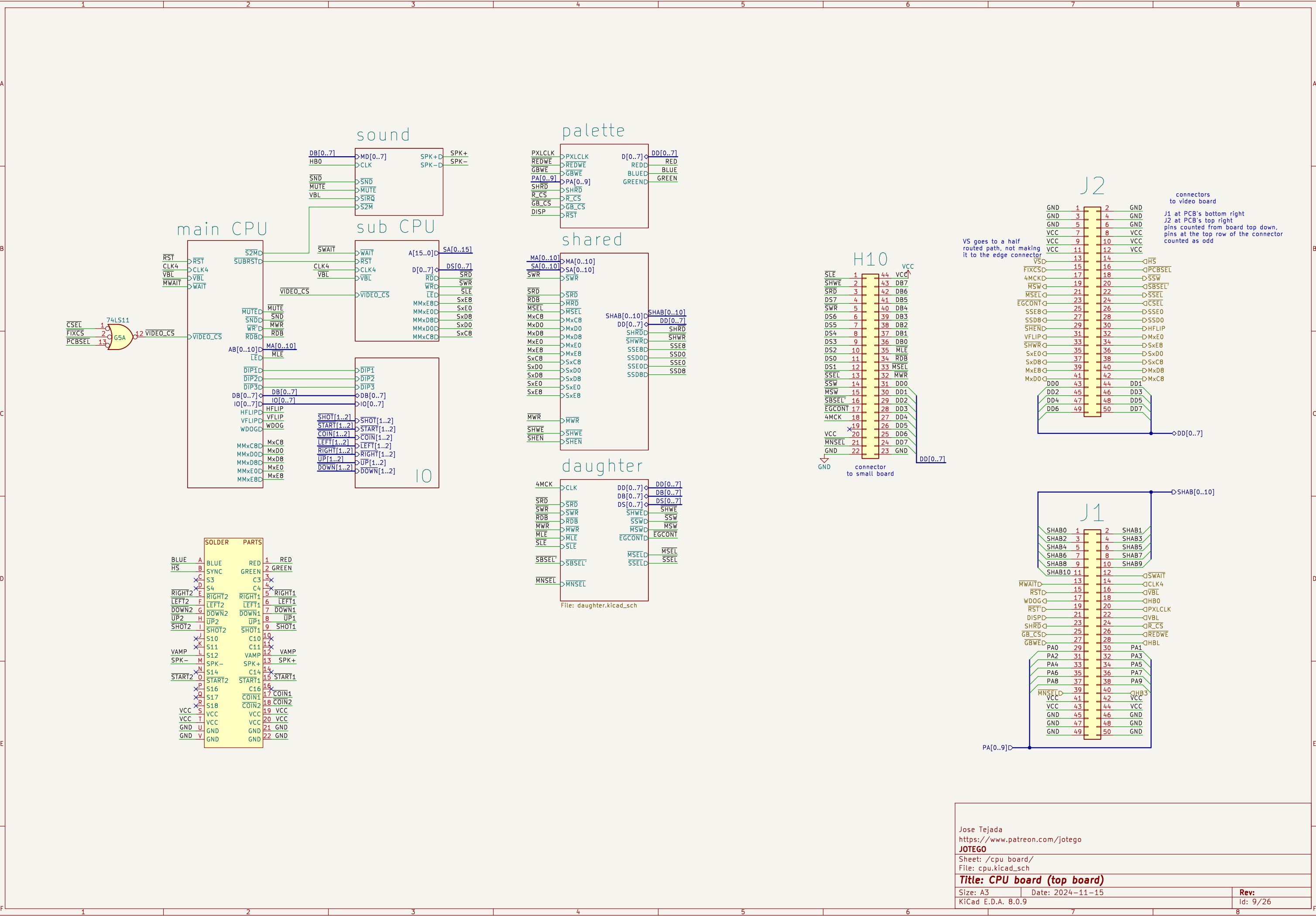


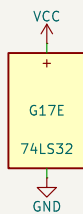
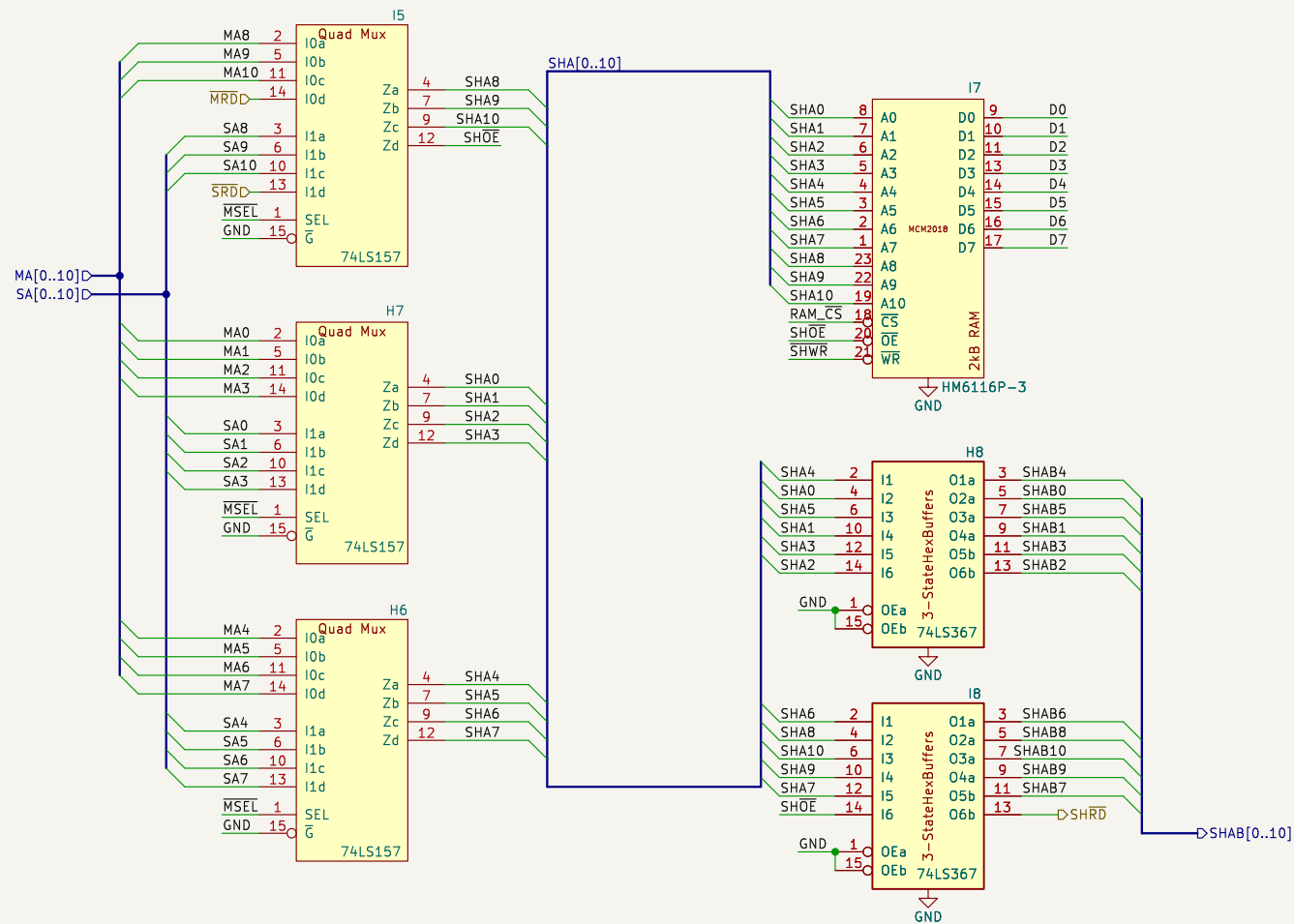
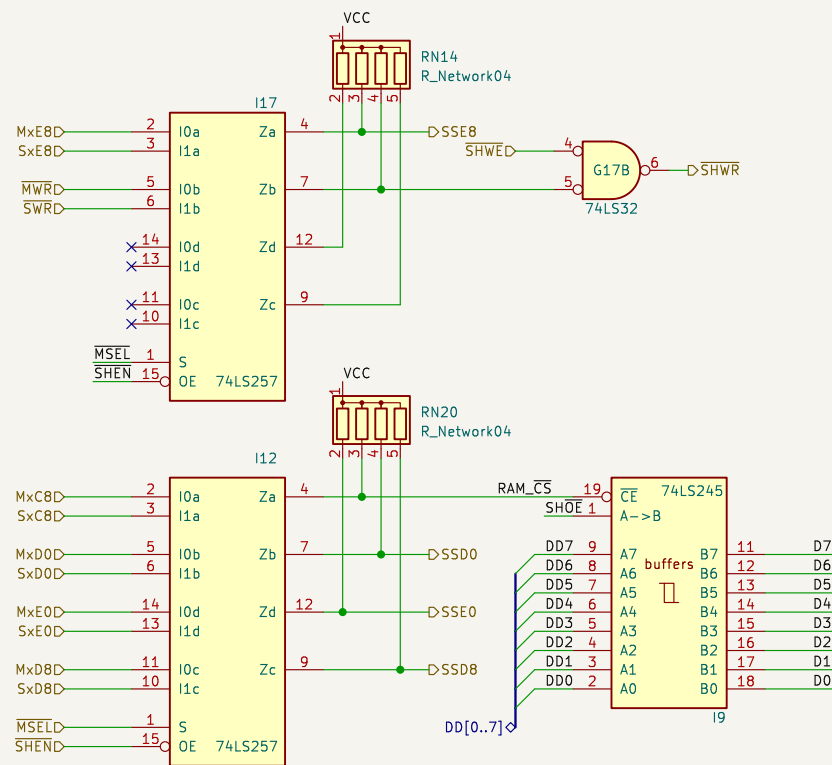
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 Sheet: /cpu_board/sound/PCM/
 File: pcm.kicad_sch

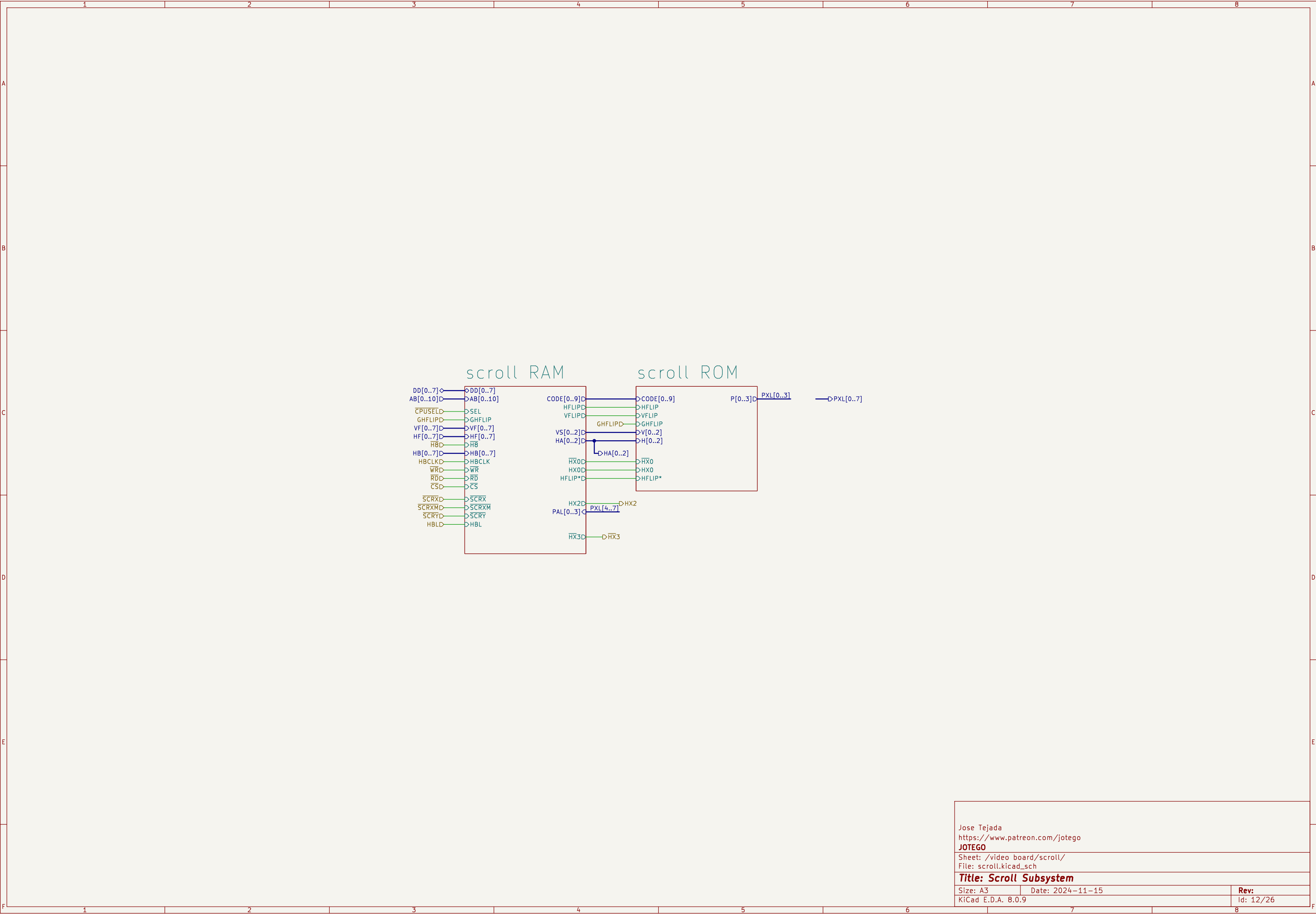
Title: PCM

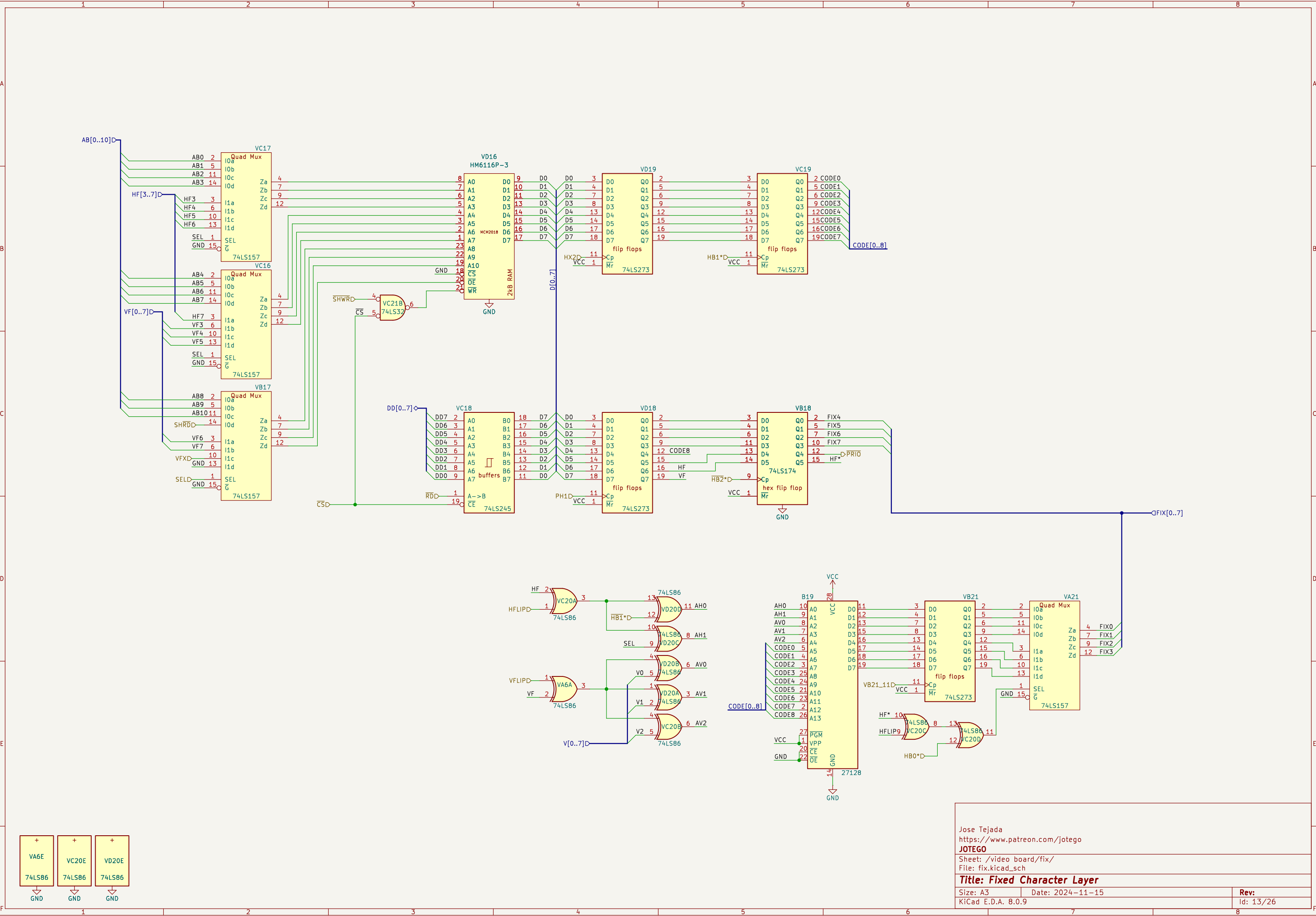
Size: A3 Date: 2024-11-15
 KiCad E.D.A. 8.0.9

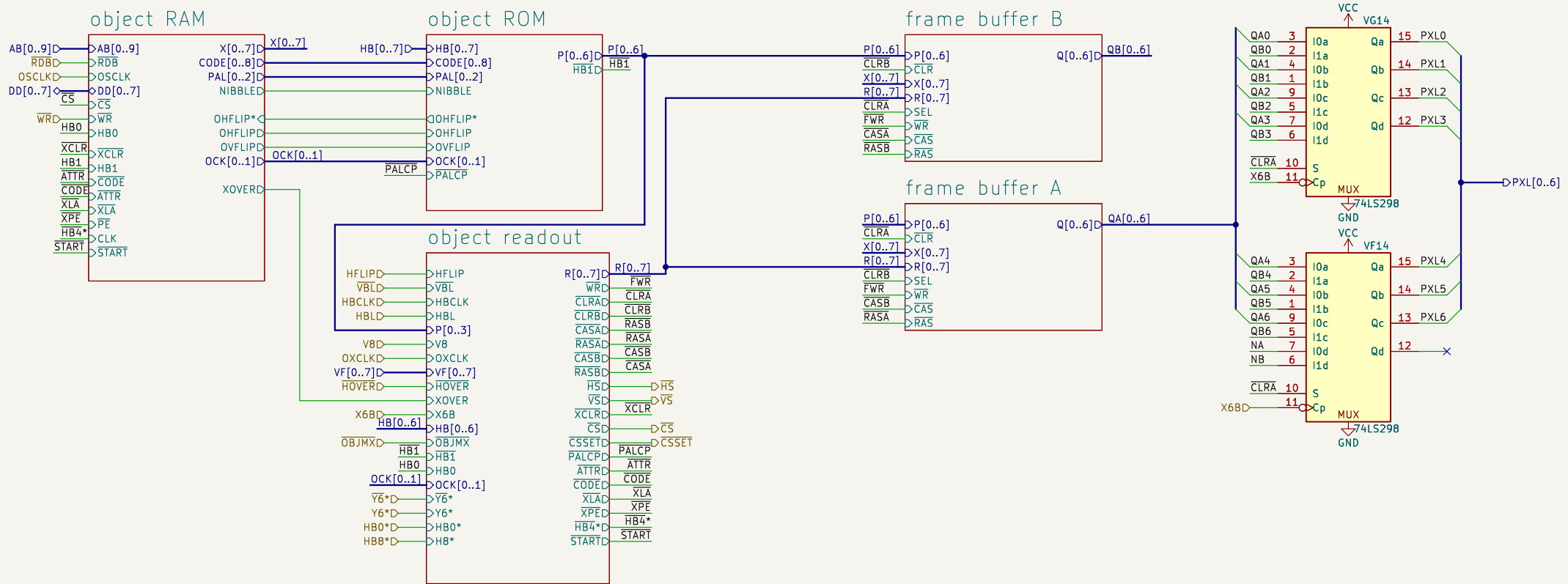
Rev:
 Id: 5/26

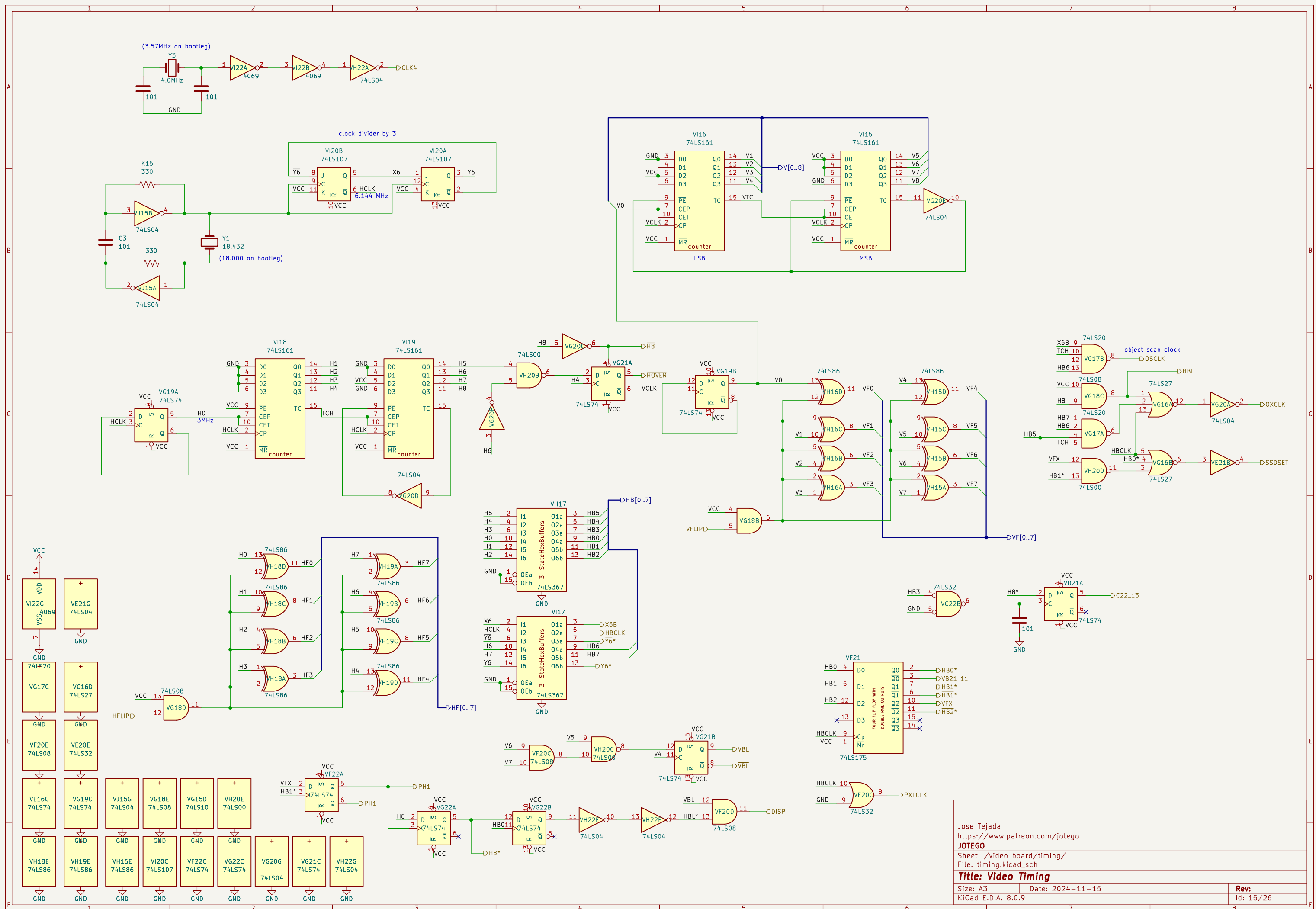


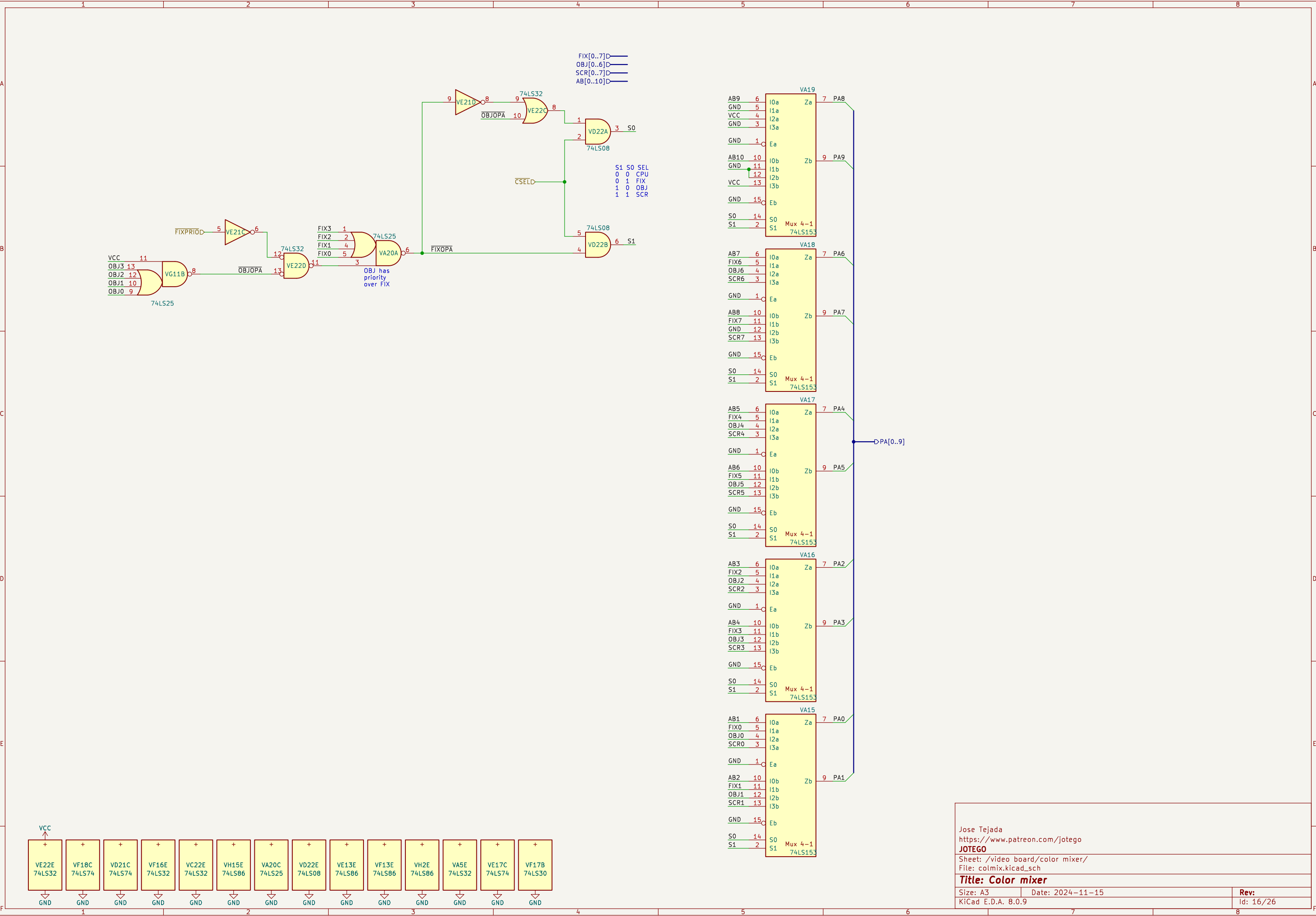


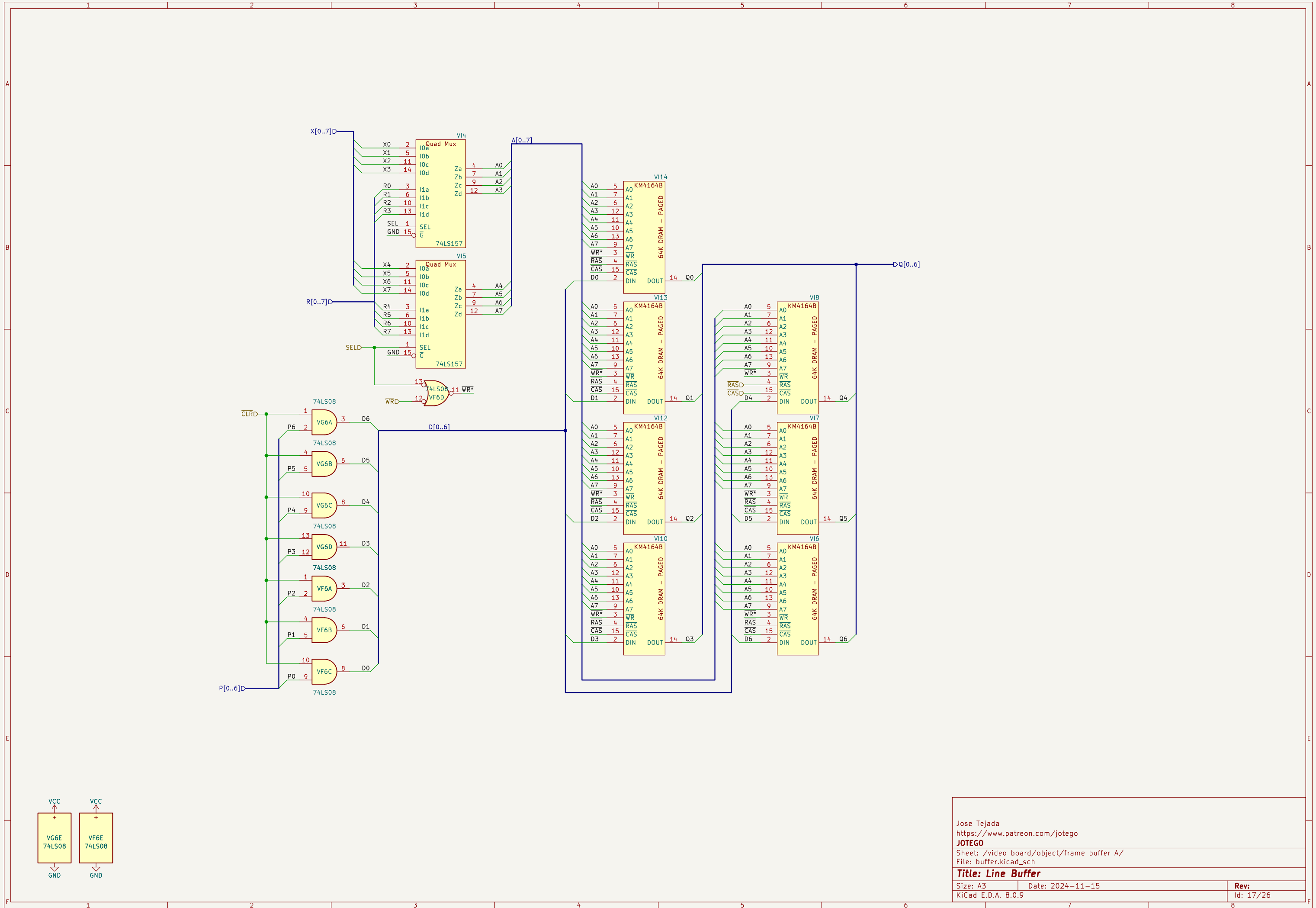












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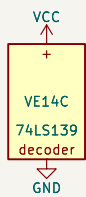
JOTEGO

Sheet: /video board/object/frame buffer A/
File: buffer.kicad_sch

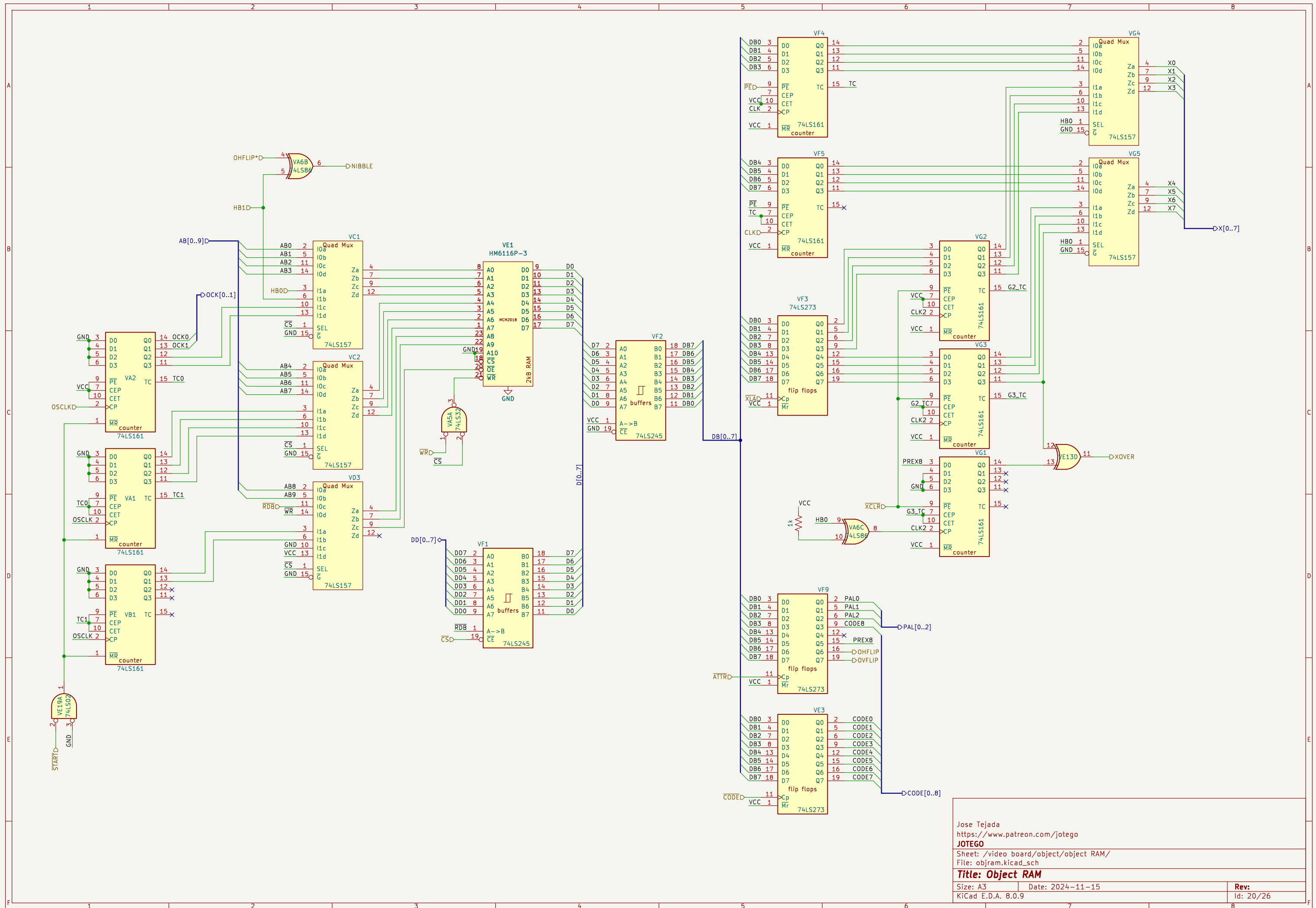
Title: Line Buffer

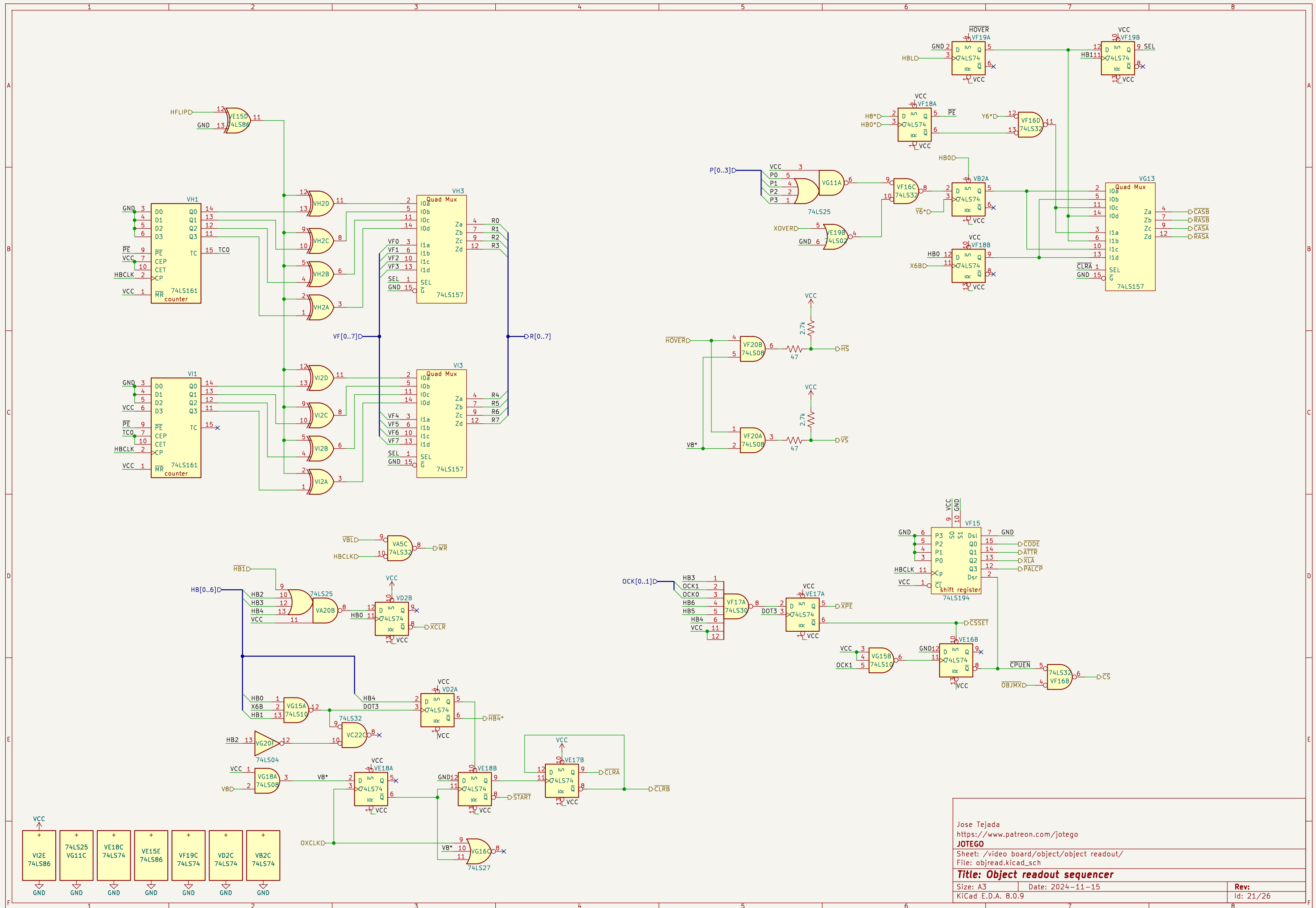
Size: A3 Date: 2024-11-15
KiCad E.D.A. 8.0.9

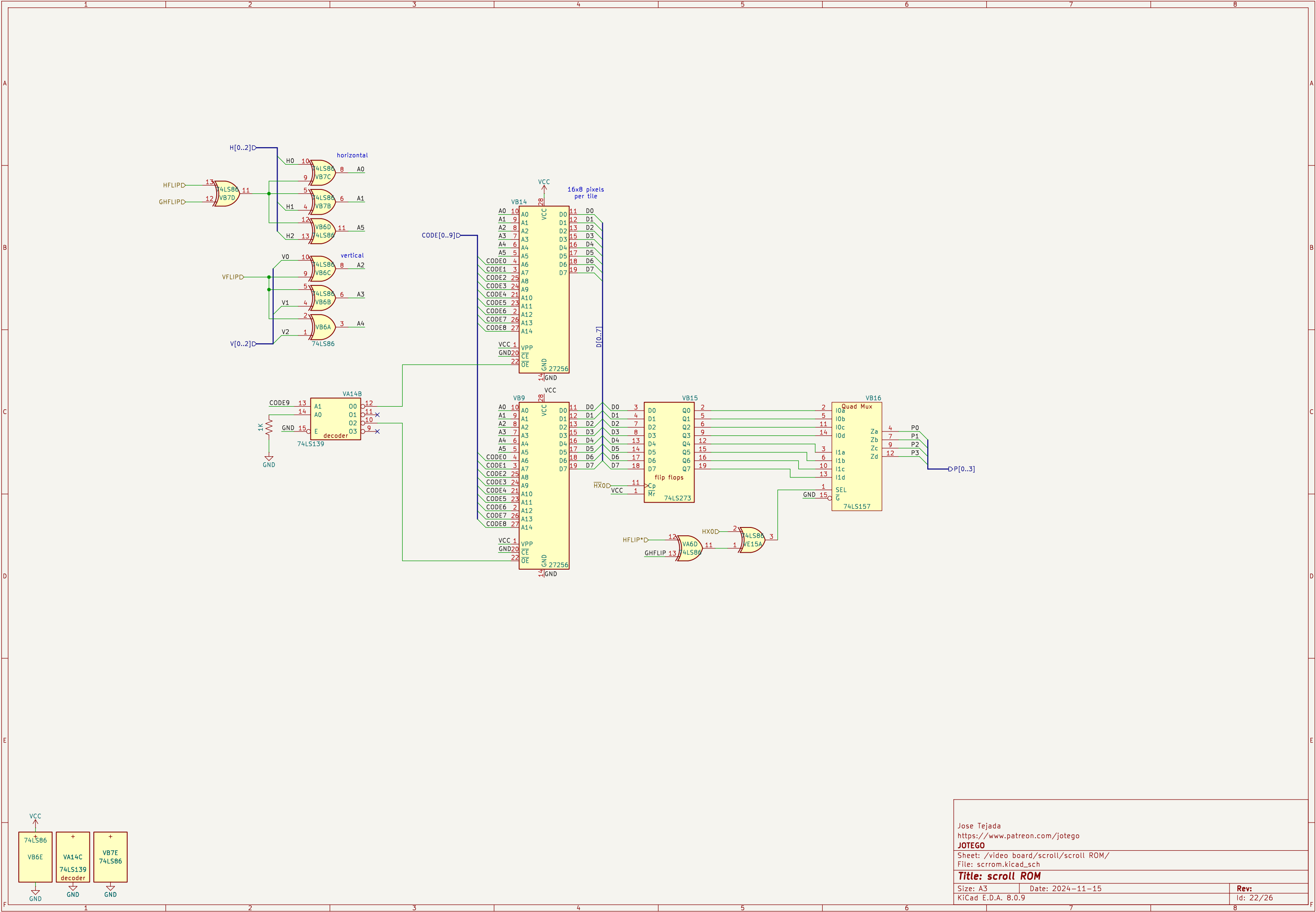
Rev:
Id: 17/26

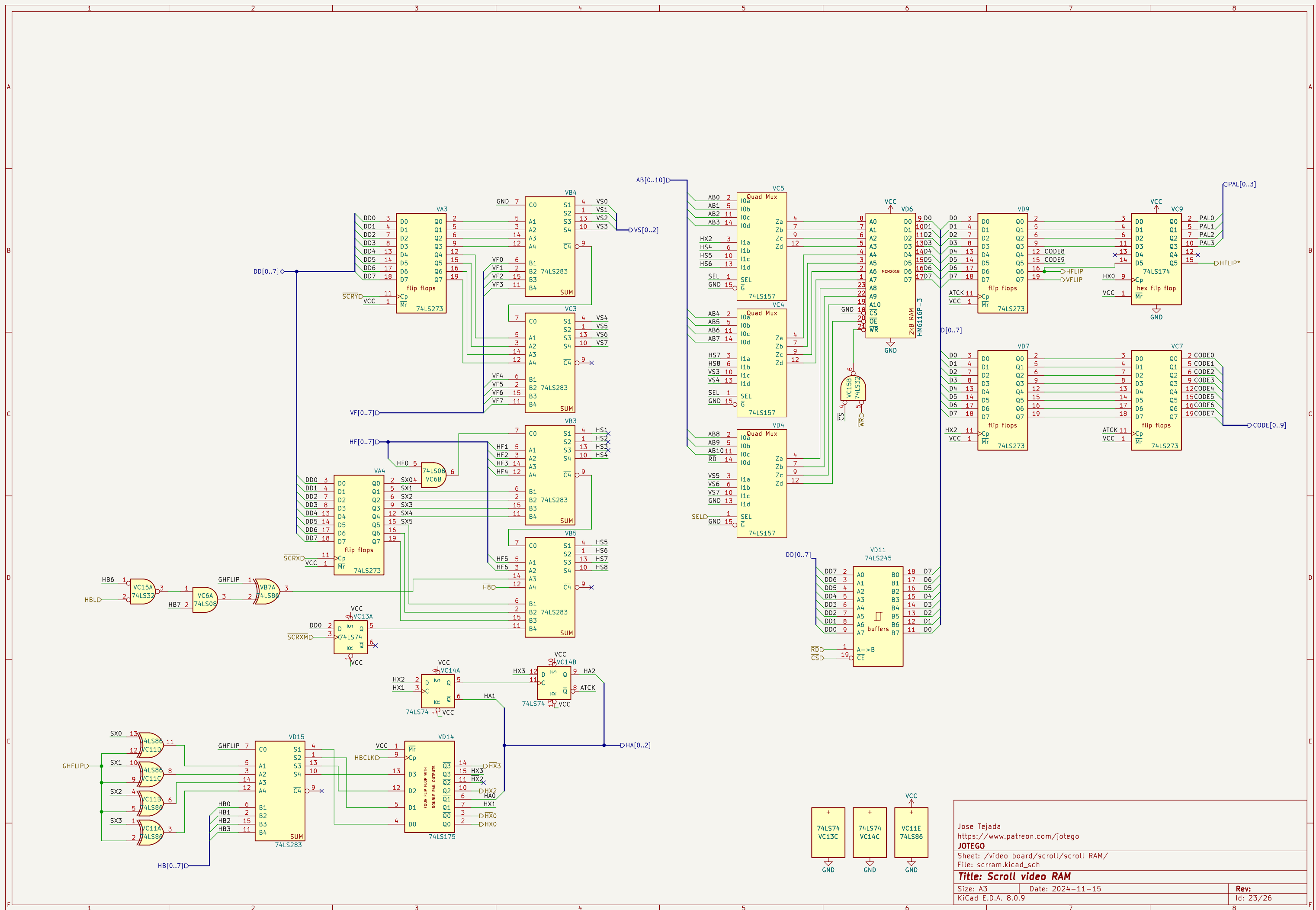


Rev:
Id: 19/26



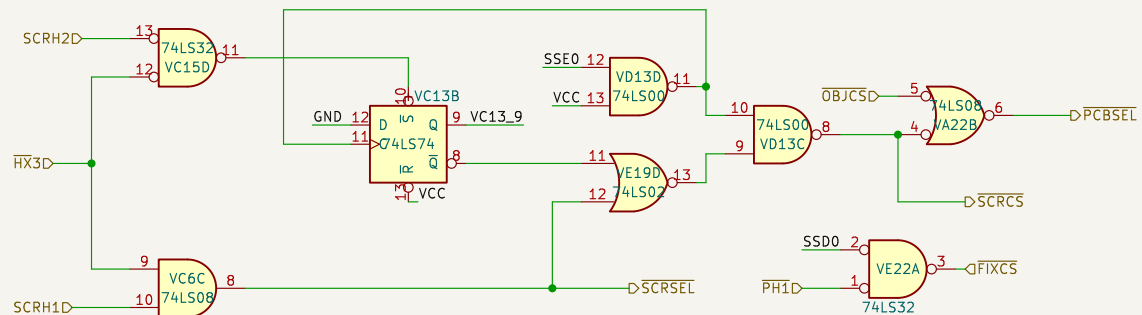


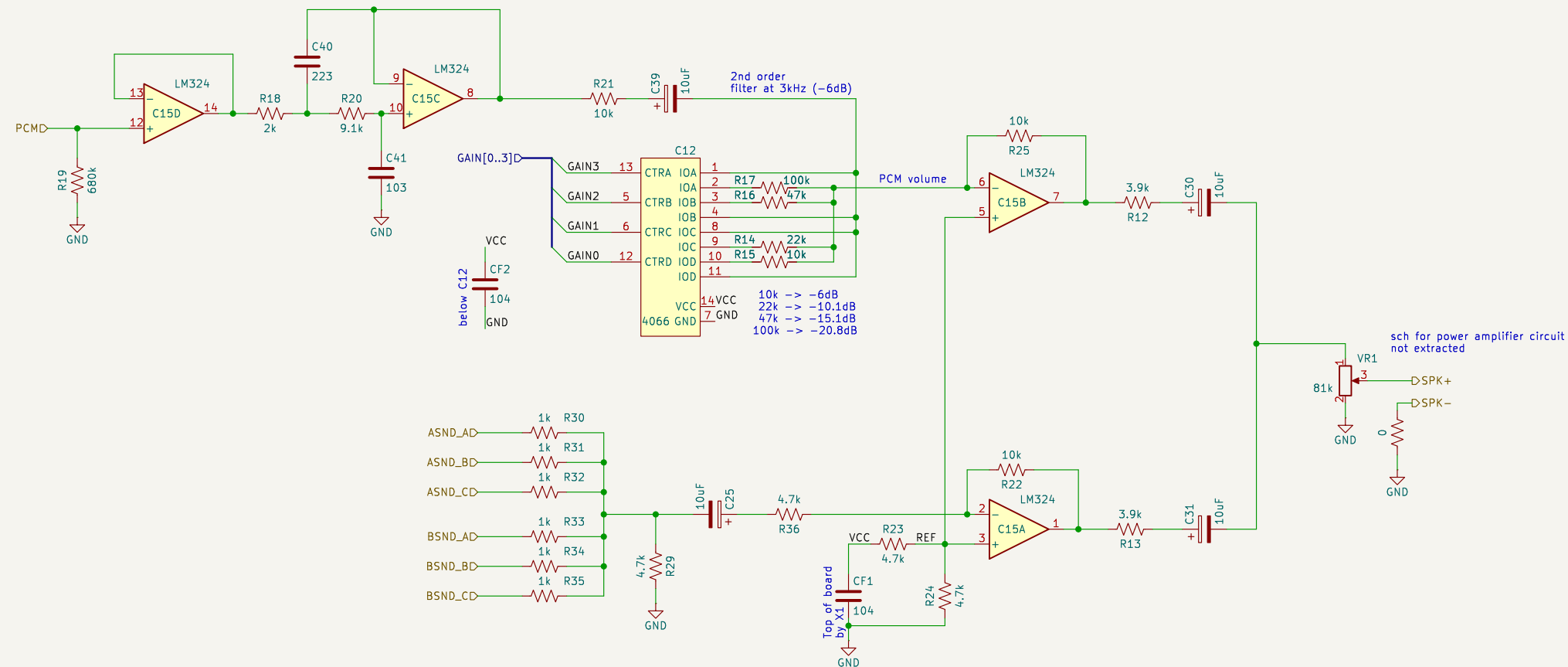






Title: Watchdog Circuit		
Size: A3	Date: 2024-11-15	Rev:
KiCad E.D.A. 8.0.9		Id: 24/26





Passive component references
when shown, come from original PCB.
The bootleg did not have
references for passives.

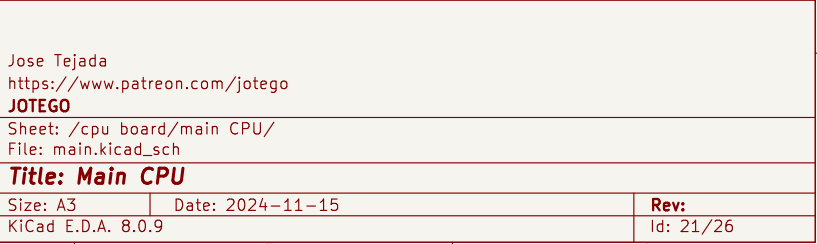
Device components follow bootleg references.
The bootleg counts letters from left to right and numbers from bottom to top
The original counts numbers left to right and letters bottom to top.

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Sheet: /cpu board/sound/analog/
File: analog.kicad_sch

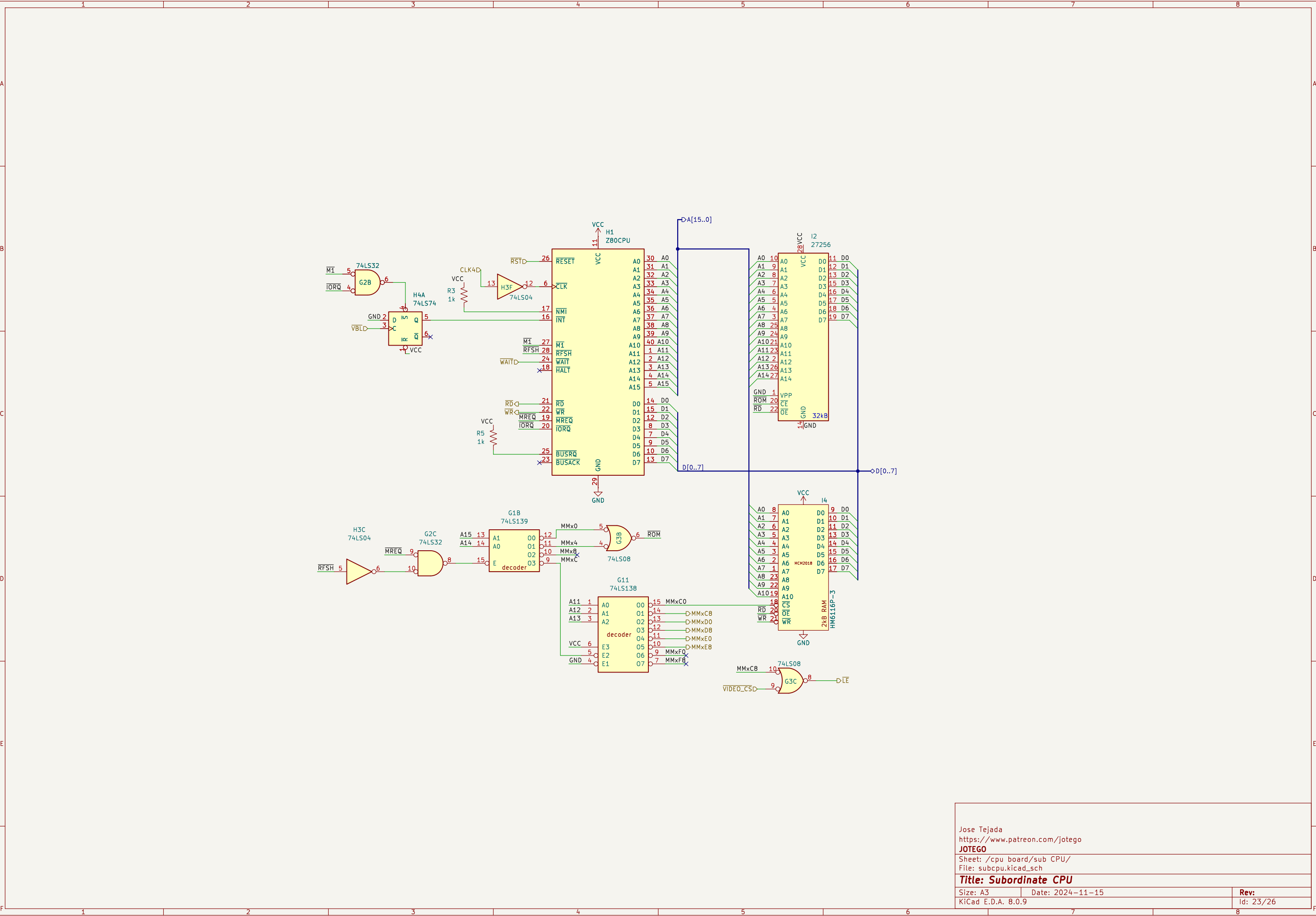
Title: Sound amplifiers

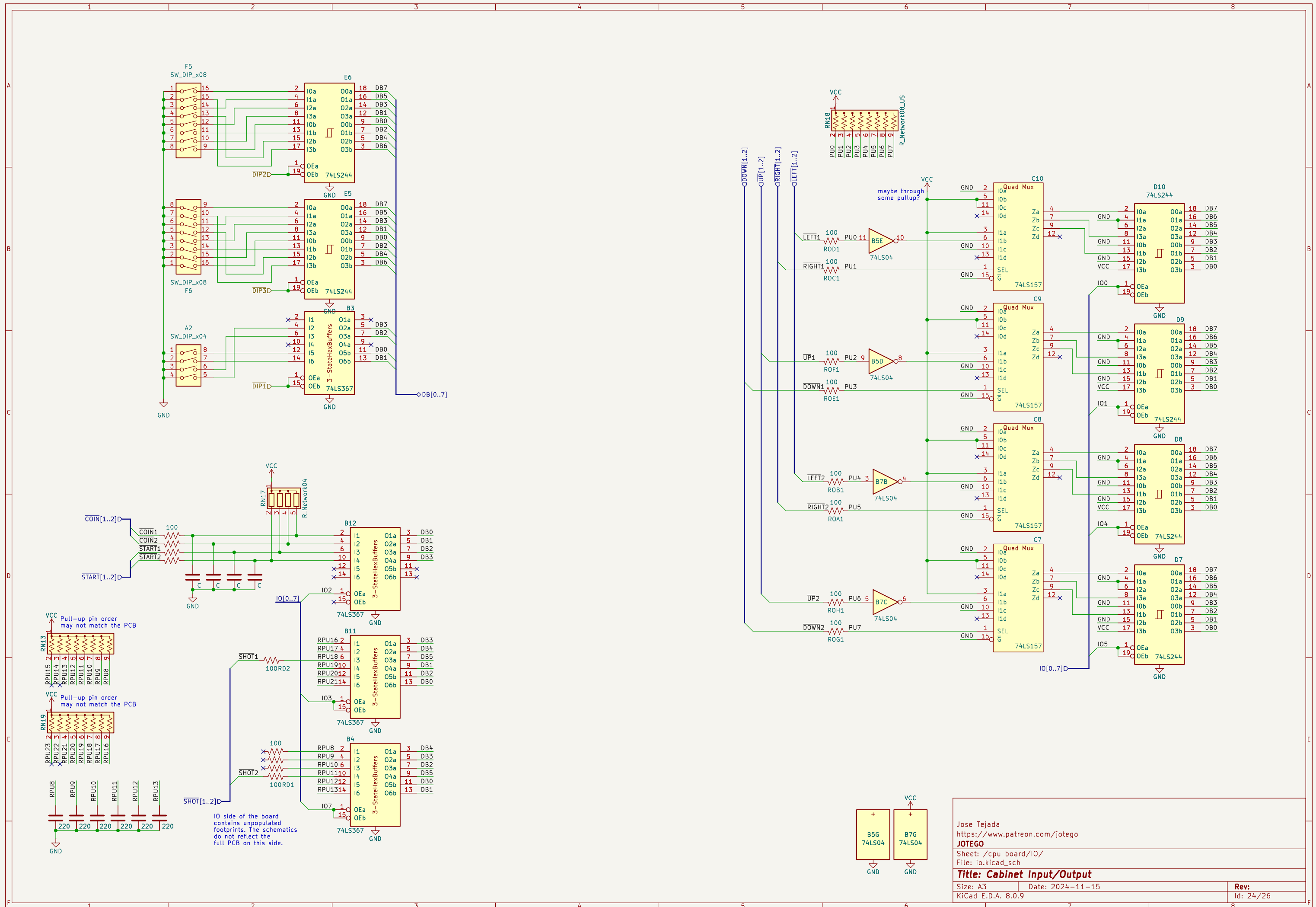
Size: A3
Date: 2024-11-15

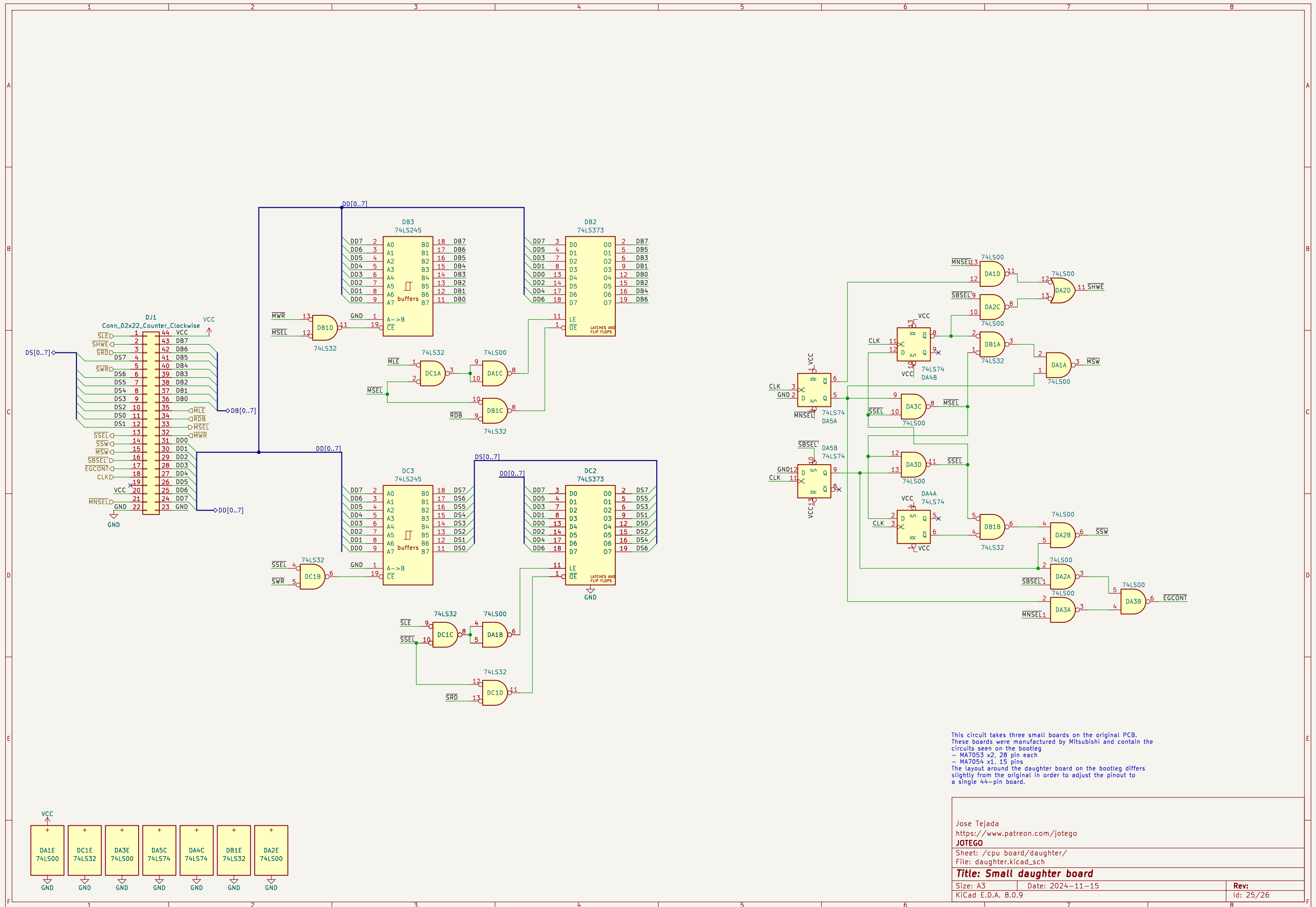
Rev:
Id: 26/26











This circuit takes three small boards on the original PCB. These boards were manufactured by Mitsubishi and contain the circuits seen on the bootleg

- MA7053 x2, 28 pin each
- MA7054 x1, 15 pins

The layout around the daughter board on the bootleg differs slightly from the original in order to adjust the pinout to a single 44-pin board.

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Sheet: /cpu board/daughter/
File: daughter.kicad_sch

Title: Small daughter board

Size: A3
KiCad E.D.A. 8.0.9

Date: 2024-11-15

Rev:
Id: 25/26

